MILITARY SPECIFICATION

COMPRESSOR UNIT, 20 CFM, 5000 PSI, DIESEL-ENGINE-DRIVEN

DIVING AIR

This specification is approved for use within the USA Belvoir Research, Development , and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a skid-mounted diesel-engine driven, air compressor for producing air suitable for breathing. Only commercial vendors approved by the Naval Sea Systems Command (CODE OOC) (NAVSEA), Alexandria, Virginia are to provide the subject unit. The Diving Equipment Authorized for Navy Use, 10 June 1987, NAVSEA Instruction 10560.2, is to be used when procuring the unit and repair/replacement parts. No variation or substitute is permitted due to the critical life support nature of the unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards , and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

BB-A-1034	- Air Compressed, for Breathing Purposes.

MILITARY

MIL-V-173	- Varnish, Moisture-and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-3600	- Compressors, Rotary, Power-Driven: and Compressors Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-G-3859	- Grease Guns, Hand Operated, Lever, Push and Screw Type.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-D-16791	- Detergent, General Purpose (Liquid Non-ionic).
MIL-L-17331	- Compressor Lubricant, Diving, Normal Temperature,
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MTL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-F-12649	- Engine, Cold Starting Aids, Ether Fuel Primers.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Service.
FED-STD-101	- Test Procedures for Packaging Materials.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	-- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-767	- Cleaning Requirements for Special Purpose Equipment, Including Piping Systems.

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MIL-STD-889	- Dissimilar Metals.
MIL-STD-1246	- Product Cleanliness Levels and Contamination Control Program.
MIL-STD-1330	- Cleaning and Testing of Oxygen and Nitrogen Gas Piping.
MIL-STD-1359	- Cleaning Methods and Procedures for Breathing Oxygen Equipment.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems.
MIL-STD-1474	- Noise Limits for Army Materiel.
MIL-STD-1622	- Cleaning of Shipboard Compressed Air Systems.
MS35844	- Plug, Machine Thread, Magnetic (Draln)-
MS39254	- Cartridge, Eng~ne-Starting (Engine Cold-Starting Aid).
MS49006	- Plug, Pipe, Magnetic-Headless, Aluminum, iron or Steel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

NAVFAC DM-39	- Hyperbaric Facilities Design Manual 39.
NAVSEA 0994-LP-001-9010	- US Navy Diving Manual Volume I.
NAVMAT P-9290	- System Certification Procedures and Criteria Manual for Deep Submergence Systems.
NAVMEDCOMINST 6260.3	- Occupational Health Surveillance Manual.
Naval Experimental Diving Unit (NEDU)	- Air and Oxygen Systems Cleaning Procedure for Organizational Level Preventative and Corrective Maintenance.
NAVSEA INSTRUCTION 10560.2	- Diving Equipment Authorized for Navy use.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR, Part 85 - Control of Air Pollution From New Motor Vehicles and New Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

FOREST SERVICE

Forest Service Standard 5100-1.

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(Application for copies should be addressed to the Forest Service, US Department of Agriculture, P.O. Box 2417, Washington, DC 20013.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Performance Test Code PTC-9 - Displacement Compressors, Vacuum Pumps and Blowers.

AMSE Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY OF TESTING MATERIALS (ASTM)

F 307 - Gas, Pressurized, Sampling, Gas Analysis, Standard Method for.

(Application for copies should be addressed to the American Society of Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW Lejune Road, P.O. Box 351040 Miami, FL 33135.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J534 - Lubrication Fittings

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rule 442 - Usage of Solvents.

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(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

STATE OF CALIFORNIA AIR RESOURCES BOARD

Health and Safety Code.

(Application for copies should be addressed to California Air Resources Board, 1709 llth Street, Sacramento, CA 95814.)

(Non-Government standards and other publications are normally avaliable from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail Specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The compressor unit shall be portable, and shall consist of a reciprocating compressor, driven by a diesel engine, and Shall produce not less than 20 cubic feet per minute (cfm) of free air at 5000 pounds per square inch gage (psig). The engine and compressor unit shall be shock mounted. The compressor unit shall be complete and self-contained, including a final air filter and all other accessories. The compressor unit shall weigh not more than 1500 pounds drv and the dimensions shall not exceed 50 inches in height, 60 inches in length, and 48 inches in width. A water resistant tarpaulin shall be provided. Diving/diving support equipment shall meet all standards of manufacturing diving/diving support equipment as approved by Naval Sea Systems Command , Code OOC, Alexandria, VA 20362.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.4). Any changes or deviations of compressors from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish compressors conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The compressor shall be fabricated from compatible materials, inherently corrosion resistant or treated

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to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the compressor may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL–STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts Incorporated in the compressor may be newly fabricated from recovered materials to the maximum extent practicable, provided the compressor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the compressor.

3.4 Environmental conditions.

3.4.1 Operation. The compressor unit shall operate as specified herein at any ambient temperature from minus 25 °F to plus 110 °F.

3.4.2 Precipitation. The compressor unit shall perform as specified herein in rain falling at a rate of three inches per hour.

3.5 Safety. All reciprocationg, rotating, or moving parts of the compressor unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the compressor unit, including exhaust pipes and other lines which may be subject to high temperatures, exposes to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Exhaust from the engine shall be directed so that it does not endanger personnel , and so that the possibility of exhaust entering into the compressor unit inlet is minimized. Electrical equipment, wires, and connectors shall be effectively guarded or insulated to protect all persons and objects from electrical shock hazard. Nonfunctional sharp edges, projecting points and excessive length of fastening devices shall be avoided.

3.5.2 Noise limits. The noise produced by the compressor shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4 when tested in accordance with 4.5.2.9. The provision of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL–STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring, activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3, except the sign will state “HEARING PROTECTION REQUIRED WITHIN (specify) FEET.” The sign shall be readable at the distance indicated on the sign.

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3.6 Reliability and maintainability.

3.6,1 Reliability. The specified mean-time-between-failure (MTBF), shall be 500 hours-d as specified in 4.5.2.14.

3.6.2 Maintainability. The compressor shall operate as specifed herein without repair of maintenance other than the contractor's recommended normal scheduled adjustments and servicing as established by a maintenance schedule prepared and submitted by the contractor prior to test. All major assemblies shall be accessible for servicing, repair, and replacement without the removal of other major assemblies. Covers and guards which must be removed for component adjustment or repair shall be equipped with quick-disconnect fasteners. All fasteners shall be of corrosion-resistant material or shall be treated to be corrosion resistant. All screw threads shall be in accordance with FED-STD-H28. Maximum use shall be made of interchangeable hardware and fasteners. Means for complete draining of lubricants, fuel and condensate traps shall be provided in an accessible location and shall be accessible for draining to a container without splashing on any component. All maintenance, assembly, or disassemble operations shall be accomplished with common tools and special tools furnished with the compressor. The maintenance ratio (MR) shall not exceed 0.05 (see 6.3.2).

3.7 Performance. The rated capacity of the compressor shall be not less than 20 cfm or free air at a delivery pressure of 5000 pounds per square inch (psi) when tested as specified in 4.5.2,16.

3.7.1 Endurance. The compressor shall operate as specified herein for 500 consecutive hours without failure. At the end of this 500 hour period, the decrease in output shall be not more than 1.5 cfm when delivering air at a pressure of 5000 psi, when tested in accordance with 4.2.13.

3.7.2 Tilted operation. The compressor shall deliver the rated capacity at 5000 psi for not less than 30 minutes when tilted in any position at an angle of 15 degrees from the horizontal. The compressor shall stand for not less than 2 hours in a tilted position without any evidence of leaks; faulty lubrication, cooling, or fuel supply and distribution; lubricant spill-over; or reduced performance upon starting after standing in this position (when tested as in 4.5.2.8).

3.8 Compressor system.

3.8.1 Air compressor pump. The air compressor pump shall be of the reciprocating design with not less than four stages. The direction of rotation shall be indicated on the unit.

3.8.2 Cooling. The compressor shall be entirely air cooled. Intercoolers shall be provided to remove the heat of compression between consecutive stages, An aftercooler shall be provided. All coolers shall have the capacity to cool the air after compression to a temperature of not more than the ambient temperature plus 25 °F in the ambient temperature range of plus 10 °F to plus 110 °F. In the ambient temperature range of plus 10 °F to minus 25 °F, positive

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control of the cooling shall be provided to prevent valves, traps, piping, and controls from freezing. If a manual device is used, a means shall be provided on the control panel to indicate to the operator the correct positioning of the device. Each cooler shall be provided with an adjustable, spring-loaded-type, pressure relief valve located between each cooler, adjusted to operate at the pressure necessary to protect the compressor.

3.8.3 Pressure tubing and fittings. All pressure tubing and fittings shall have a minimum working pressure of 5000 psi and a test pressure of 8300 psi, and shall be of stainless steel type 304 or equivalent. All tubing and fittings shall be installed to minimize and withstand vibration and allow for thermal expansion. Provision shall be made for relieving pressure to the atmosphere from all lines in the system without disassembly of any fittings or component.

3.8,4 Valves. Manual operated shutoff, isolation, and control service valves shall be of the positive shutoff type and shall be of stanless steel type 304 or equivalent . All manually operated valve controls shall be provided with double-ended arrows showing the direction of operations and labled at each end to indicate the functional result. The valves shall require not more than 60 inch-pounds of torque to open or close under a pressure of 5000 psi and overtorquing to 150 inch-pounds shall not damage the valve seats. Valve seats shall be resistant to seat erosion.

3.8.5 Moisture separator. A mechanical-type moisture separator shall be provided downstream from the aftercooler to remove condensed moisture and oil. The separator shall have a minimum operating pressure of 5000 psi and a proof pressure of 8300 psi, and shall be of corro~iort-resistant material. The separator shall have a volumetric capacity large enough to contain the moisture and oil which would accumulate in 20 minutes while the compressor is operating at rated capacity with a relative humidity of not less than 85 percent and an ambient temperature of 110 °F. A valve shall be provided for periodic drainage of the moisture separator. Periodic drainage shall not be required more frequently than 15 minute intervals under all conditions specified herein. In addition, automatic drainage of the moisture separator shall be provided on shutdown of the compressor. Provisions shall be made to heat the moisture separator when operating at ambient temperature below 32 °F to prevent freezing of condensed moisture. When an electrical heater blanket is utilized, it shall be for 28-volt dc electric service.

3.9 Comerical engine.

3.9.1 Commerical engine. For the purpose of selecting the engine, the compressor shall be classified as a class II end item in accordance with MIL-STD-141O. The engine, including all systems, components, and accessories, shall conform to MIL-STD-141O, except as otherwise specified herein. The engine shall be 4-cycle and air cooled. The engine shall be furnished as a complete, self-contained, power package including hand throttle control, fail-safe governor, manual starting system, and engine stop switch. The engine shall comply with the Environmental Protection Agency and the California Air Resources Board regulations governing control of air pollution for new motor vehicles and new engines, in effect for this type of equipment on date of manufacture. The

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engine shall be capable of operating under all conditions specified herein when using fuels and lubricants specified herein and in MIL-STD-1410, without exceeding any of the operating limits specified herein and in table I of MIL-STD-141O, or the engine manufacturer's recommended operating limits. An auxillary priming system, conforming to MIL-E-52649, type III and MS39254 shall be furnished.

3.9.2 Fuel tank. As specified, a fuel tank with a capacity for not less then 8 hours of operation at rated load shall be provided (when tested in accordance with 4.5.2.13).

3.9.3 Drive. The compressor shall be either direct or multiple V-belt drive. When a direct drive is used, a flexible coupling shall be installed between the power unit and compressor unit. The power unit shall have a means for belt removal and adjustment when V-belt drive is used.

3.9.4 Exhaust system. The exhaust system shall be designed to present minimum hazard to the operator while starting, operating, or making minor adjustments to the engine or compressor unit while running. The system shall include a muffler and a spark arrestor, or a combination muffler and spark arrestor. The spark arrestor shall conform to the US Department of Agriculture, Forest Service Standard 5100-1. The total backpressure of the exhaust system shall not exceed the maximum recommended by the engine manufacturer. The complete exhaust system shall be protected from the entry of rain,

3.10 Air systems.

3.10.1 Air induction system. A flexible noncorrosive hose of proper diameter shall be provided as an air intake extension. The hose shall not be less than 15 feet in length and shall have an adapter at one end to attach to the intake of the compressor air cleaner while the opposite end is laying on the ground. Provisions shall be made for storing the hose inside the compressor frame when the compressor is not in use. A means for preventing foreign objects from entering the flexible hose shall be provided.

3.10.2 Cooling. Cooling shall be provided by a fan or fans directing an air blast over the compressor, intercoolers(s), and aftercooler(s). The inter-cooler(s) shall be furnished to cool the air between stages and each shall be equipped with a safety valve to protect the low pressue cylinder in case of failure of the high pressure cylinder valves. The aftercooler(s) shall be furnished between the compressor air discharge and final filter. The air temperature measured at the inlet of the final filter shall not exceed 135 °F under all operating conditions specified herein.

3.10.3 Final filter. The final filter shall be mounted vertically and installed downstream of all other pneumatic system components, and shall be designed for a working pressure of 5000 psi and a test pressure of 8300 psi. With the compressor delivering air at rated capacity, under all operating conditions specified herein, the filter element shall provide air conforming to the purity requirements of BB-A-1034, grade C, ASTM F 307, and NAVSEA 0994-LP-001-9010 for a period of not less than 80 hours, except that the total water

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limit shall not apply. A moisture separator shall be installed upstream of the final filter to prevent liquid water from reaching the filter element. The design of the filter housing shall provide for replacement of the filter element by one man using only common tools. Gaskets and gasket sealer shall be nontoxic material. The final filter shall have a carbon monoxide removal capability and the delivered air shall not contain more than 10 parts per million cabon monoxide when 50 parts per million cabon monoxide are introduced into the compressor intake in accordance with air purity standards found in NAVSEA 0994-LP-001-9010. Filtration system and filter elements will be standardized where possible and limited to those filtration systems and filter elements on the "Approved for Navy Use List" (NAVSEAINST 10560.2).

3.11 Pressure control valve. A pneumatic pressure control valve shall be provided to prevent the passage of air through the filter bank at pressure less than 2000 psi. The unit shall be equipped with a check valve to prevent reverse air flow of downstream air.

3.12 Control panel. A control panel shall be provided for mounting any gages, meters, and controls as specified herein. Gages, meters , and controls shall be clearly identified and shall be mounted so that they will not be damaged under any condition of compressor operation specified herein. All gages, controls, and meters necessary for operation shall be accessible and visible to a single operator standing on one side of the compressor. A schematic airflow diagram shall be permanently affixed on or near the control panel. Labeling shall comply with requirements in 3.19.2.

3.12.1 Instrumentation. Pressure gages shall be provided to indicate filter system and servicing line pressures. All pressure gages shall be the liquid fill type, mounted on the control panel and shall be isolated from vibration by shock absorbing mounts. Each gage shall be vibration resistant and provisions shall be made to prevent freezing of gases under conditions specified.

3.12.2 Hourmeter. An hourmeter shall be mounted on the control panel. The hourmeter shall have a range from 0 to 9999 hours and shall be connected to tne compressor.

3.12.3 Filler hose. A filler hose shall be provided to connect the compressor and diving tanks so that the tanks may be filled. The hose shall be a commertial type hose, 15 feet or longer In length with a minimum 60O0 psi working pressure and a 1(.)000 psi test pressure. The hose shall have a wire-bra~a outer layer and an innermost layer of polytetrofluoroethylene. The hose shall have SCUBA Industry standard quick release/connect fittings at each end of the nose, an approved military diving pressure gage and a SCUBA fill yoke complete with integral fill valve and bleeder valve.

3.13 Skid mounting. The compressor unit shall be assembled and mounted on a skid. Both ends of the skid frame shall be provided with pulling attachments which may also be used for lifting or tiedown purposes. The skid frame shall include two channels not less than 11 inches wide and 3 inches high opening on both sides of the compressor unit to provide for moving by forklift. The center distance between the channels shall be not more than 64 inches.

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3.13.1 Frame. A frame, mounted on the skid, shall be provided around the compressor unit. The slinging and tiedown provisions (see 3.17) shall be mounted on the frame as high as practical. The frame shall be designed to support tile gross. weight of the compressor unit (including skid and frame) distributed on a flat plane from any side of the frame or on the top. The skid shall be within the dimensions stated in 3.1.

3.14 Electromagnetic Interference. The electromagnetic interference emission characteristics of the compressor unit shall conform to MIL-STD-461, class III C.

3.15 Lubrication. All surfaces requiring lubrication shall be provided with a means of lubrication. Only approved lubricants as specified in NAVSEA 0994-LP-001-9010 will be used on the life support portions of the compressor unit.

3.15.1 Lubricants. The compressor unit shall operate as specified herein when lubricated with military lubricants, conforming to MIL-L-17331 (2190 TEP), normal operations. No substitutions will be allowed,

3.15.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859 with 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.15.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.15.4 Filling, draining and checking provisions. Enclosures that contain a reservoir of lubricants for the lubrication of the parts enclosed, shall be equipped with dipsticks, check plugs, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a drain plug, size conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of lubricant from the enclosure. Lubricants shall be capable of draining to a container when the compressor 16 in its normal position. Integral tubes or troughs may be used to convey Lubrication from the drain outlet.

3.15.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-2387 - For ball, roller and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments, electronic gear, and mechanical control systems; for use on gears, actuator screws and other equipment requiring a lubricant with high-load carrying capacity over a temperature range of minus 100 °F to plus 250 °F.

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b. MIL-G-1094 - For lubrication and surface corrosion protection of all automotive equipment operated over the temperature range of minus 65 ‘F to plus 225 °F,

The compressor unit shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases may- not be compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.15.6 Oil lubrication. Unless otherwise specifed herein, all component shall be filled to the operating level with military oils designated for use in the temperature range specified herein. A tag shall be attached in a conspicuous location to indicate the temperature range grade of lubricant used, and frequency of lubrication. A warning sign shall be attached to the frame of the compressor stating “Do not mix or interchange compressor lubricants with engine lubricants. Use only MIL-L-17331 (2190 TEP) lubricant for compressor lubrication.”

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.16.1. With 1 percent copper-8-qulnolinolate (by weight) based on the nonvolatile content of the varnish, except:

a. Components of circuit elements that are inherently fungus and moisture resistant or which are heretically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition, Composition I shall be used unless composition II must be used in order to comply with local air pollution regulations In the application of the varnish When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442 - South Coast Air Quality Management District.

3.17 Slinging and tiedown provisions. Permanently affixed attachment for lifting and tying down the compressor unit shall be provided, and conform to MIL-STD-209 for Type IV equipment, including the requirements for air transport. Slinging and tiedown provisions shall be labeled “LIFT HERE” and “TIEDOWN” respectively.

3.18 Transportability. The compressor unit shall withstand the stresses encountered in shipment including those of the pendulum-impact test specified in 4.6.2.17 without deformation or damage which results in de.gradation of compressor unit performance.

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3.19 Identification marking. The compressor unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the compressor on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class I material. Plates shall be attached by screws, bolts or rivets in a conspicuous protected location.

3.19.1 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class material shall be provided. The plate shall also show the silhouette of the compressor unit in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. Plates shall be attached by screw, bolts, or rivets in a conspicuous protected location.

3.19.2 Instruction plates and labeling. Each compressor unit shall be equipped with instruction plates, including diagrams, warnings, and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the compressor unit. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class I material. Plates shall be attached by screws, bolts or rivets in a conspicuous protected location. Labeling shall comply with MIL-STD-1472, 5.5.

3.20 Treatment and painting. The portions of the compressor unit normally painted internally and externally shall be cleaned, treated, and painted in accordance with NAVFAC DM 39, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.21 Workmanship.

3.21.1 Casting and forgings. All parts, components , and assemblies of the compressor unit which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.21.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by method that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel q ay be employed instead of shearing and sawing, All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.21.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing and other foreign matter. Welds shall transmit stress without pemananent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the compressor unit to conform to the examination and test requirements specified in section 4. Parts to be jointed by fillet welds shall be brought into as close contact as possible and in no event

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shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the compressor unit shall be at the option of the contractor.

3.21.4 Welders. Before assining any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ASME Boiler and Pressure Vessel Code, Section IX, “Welding and Brazing Qualifications.

AWS, D1.1 Structural Welding Code - Steel.

Contractors who only make horizontal welds need not qualify welder for “all position welding”. The government reserves the right to require recertification of any welder or welding operator. The certification shall be made available for review by the contracting officer or his representative upon request.

3.21.5 Bolted connections. Boltholes shall be accurately formed and shall have all burrs removed. Stainless steel washers shall be provided where necessary. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.21.6 Riveted connections. Rivets shall fill the hole completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.21.7 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and to the standards of the industry.

3.22 Weight. The completed unit gross weight shall not exceed 1500 pounds dry weight.

3.23 Human factors engineering. The compressor unit shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety) of MIL-STD-1472, as applicable.

3.24 Cleaning of life support parts and systems. All parts and systems contained with the unit for use with oxygen and compressed air will be cleaned and tested for cleanliness in accordance with the standards of the following publications and procedures: MIL-STD-767, MIL-STD-1246, MIL-STD-1330, MIL-STD-1359, MIL-STD-1622, MIL-D-16791, NAVMAT ?-9290, NAVMEDCOMINST 6260.3, and NEDU Air and Oxygen Systems Cleaning Procedure for Organizational Level Preventative and Corrective Maintenance. Included in the cleaning procedure is

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the requirement to perform an air purity sampling test after the last step of the cleaning procedure is completed to insure that the parts and systems cleaned meet the exposure limit values for breathing gas as stated in NAVMEDCOHINST 6260.3.

3.25 Conflicts. If a conflict arises with regard to which procedure or publication to use, the contracting officer will be notified in writing by the contractor prior to any work being done (see 6.7).

4. QUALITY ASSURANCE PROVISONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material,

4,1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. inspections shall be classified as follows:

a. First-article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. Prior to testing, the first article compressor unit shall be examined for defects as specified in table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article compressor unit shall be subjected to the tests marked "X" in column 1 of table II. Failure of any test shall be cause for rejection. A test log shall be kept recording all appropriate test data. A copy of the test log shall be included in the test report.

4.4 Quality conformance inspection.

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4.4.1 Examination. Each compressor unit shall be examained for the defects as specified in table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each compressor unit shall be subjected to the tests marked "X" in column 2 of table II. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination.

Table I. Examination schedule.

Defects 1	Requirements 2
101. Compressor unit not as specified.	3.1
102. Weight or dimensions not as specified.	3.1
103. Configuration not as specified.	3.1
104. Materials not as specified.	3.3
105. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
107. Contractor does not have documentation available for identification of material, material finishes, or treatments.	3.3.3
108. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the compressor.	3.3.4
109. Safety provisions not as specified.	3.5
110. Maintainability provisions not as specified.	3.6.2
111. Direction of rotation not indicated.	3.8.1
112. Cooling not as specified.	3.8.2
113. Tubing not as specified.	3.8.3
114. Valves not as specified.	3.8.4, 3.11
115. Moisture separator not as specified.	3.8.5
116. Engine or accessories missing or not as specified.	3.9, 3.10
117. V-belt drive not as specified.	3.9.3
118. Exhaust system not as specified.	3.9.4
119. Air induction system not as specified.	3.10
120. Final filter not as specified.	3.10.3
121. Instruments and controls missing or not as specified.	3.12
122. Skid not as specified.	3.13
123. Frame not as specified.	3.13.1

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Table I. Examination schedule (Cont'd)

Defects 1	Requirements 2
124. Lubrication provisions or tags missing or not as specified.	3.15 thru 3.15.6
125. Fungus and moisture resistance not as specified.	3.16
126. Slinging and tiedown provisions not as specified.	3.17, 3.13.1
127. Identification marking missing or not as specified.	3.19
128. Shipping data plates or instruction plates missing or not as specified.	3.19.1, 3.19.2
129. Treatment and painting not as specified.	3.20
130. Paint color not as specified.	3.20
131. Workmanship not as specified.	3.21 thru 3.21.7
132. Welders' certification not made available for review.	3.21.4
133. Weight not as specified.	3.22
134. Human factors engineering not as specified.	3.23
135. Cleaning not as specified.	3.24

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the compressor shall be lubricated with military oils and greases, 0ils shall be those designated for use in ambient temperature at place of test. The compressor unit shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. Unless otherwise specified herein, tests shall be conducted with all compressor unit components and accessories, including the air intake extension, installed and functioning normally. Capacity shall be determined in accordance with ASME Performance Test Code (PTC-9) or by calculating capacity based on a weight of air delivered in a given time. Condensate removed from the air shall not be included as part of the compressor capacity. If the weight method is used, the reservoir (minimum volume 1000 cubic inches) shall be filled to 3200 plus or minus 100 psig, and the capacity shall be deter-mined by the following formula:

$$CFM = \frac{W2 - W1}{.0764} \times \frac{R}{520} \times \frac{1}{t} \times \frac{29.92}{P}$$

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where CFM = Cubic feet per minute (free air)
W1 = Empty weight of capacity check assembly (lb)
W2 = Full weight of capacity check assembly (lb)
R = Air temperature at cooling fan in degrees, Rankine (Fahrenheit, absolute)
t = time required to fill reservoir in minutes
P = Barometric pressure in inches, Hg

4.5.2.2 Test schedule. Tests shall be in accordance with table II. It is not required that tests be performed in the order listed, except that the operation test shall be the last test performed.

Table II. Test schedule.

First article	Individual	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	X	Test conditions.	4.5.2.1	3.8
X	-	High temperature.	4.5.2.5	3.4.1, 3.8, 3.10.2
X	-	Low temparature.	4.5.2.6	3.4.1
X	-	Precipitation.	4.5.2.7	3.4.2
X	-	Tilted position.	4.5.2.8	3.7.2
X	-	Sound level.	4.5.2.9	3.5.1
X	X	Air Purity	4.5.2.11	3.10.2
X	-	Electromagnetic interference.	4.5.2.11	3.14
X	-	Slinging and tiedown provisions.	4.5.2.12	3.17
X	-	Endurance.	4.5.2.13	3.71
X	-	Reliability.	4.5.2.14	3.6.,3.6.1
X	X	Maintainability.	4.5.2.15	3.6.2
X	X	Operation.	4.5.2.16	3.7
X	-	Pendulum.	4.5.2.17	3.8

4.5.2.3 Test log. A test log covering the run-in, engine endurance, cycling, high and low temperature, tilted operation, cleaning procedures and air purity test results shall be established tabulating the appropriate test data. Storage temperature , precipitation, lifting and tiedown, pulling, towing, and pendulum impact test as applicable, shall also be recorded in this log.

4.5.2.4 Test condtions.

a. Unless otherwise specified herein, sequence of tests is optional with the tester.
b. Only that maintenance established by the contractor and submitted as maintenance schedule prior to start of testing shall be perfomed during the test.
c. Safety and human factor considerations shall be evaluated throughout testing.

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4.5.2,5 High temperature.

4.5,2.5.1 Test procedure. Subject the compressor unit to an ambient temperature of plus 110 °F, plus or minus 5 °F until the fuel, and the engine and compressor unit lubricating oil temperatures are stabilized. Start and operate the compressor unit at rated speed and capacity for not less than 8 hours while maintaining the 10 °F plus or minus 5 °F ambient temperature. No adjusting or replacing of engine or compressor components shall be permitted after the first hour of operation.

4.5.2. 5.2 Test data. Record the following data at 30 minute intervals:

a. Ambient air temperature in test cell, 3 to 5 feet from the unit at crankshaft level.
b. Engine lubricating oil temperature (gallery).
c. Compressor lubricating oil temperature (sump).
d. Compressor discharge temperature (measured at inlet to final filter).
e. Compressor air pressure at discharge.
f. Compressor capacity.

4.5.2.5.3 Failure criteria. Any one or more of the following shall constitute failure of this test.

a. Engine cylinder head temperature higher than the maximum specified 3.9.
b. Engine lubricating oil temperature higher than the maximum specified in 3.9 for the lubrication system.
c. Compressor air discharge temperature higher than 135 °F.
d. Compressor air delivery pressure less than 5000 psig.
e. Compressor discharge capacity less than 20 cfm.

4.5.2.6 Low temperature.

4.5.2.6.1 Test procedure. Subject the compressor unit to an ambient temperature of minus 25 °F plus or minus 5 °F until the fuel, and the engine compressor lubricating oil temperatures are stabilized. Start and operate the compressor unit at rated speed and capacity for not less then 4 hours while maintaining the minus 25 °F plus or minus 5 °F ambient temperature. Test data shall be recorded as specified in 4.5.2.5.2.

4.5.2.6.2 Failure criteria. Any one or more of the following shall constitute failure of this test.

a. Inability of the compressor unit to start without the use of aids, other than the engine priming system.
b. Stabilized engine lubricating oil temperature not within the limits specified in 3.9 with the engine operating.
c. Compressor air discharge temperature higher than 135 ‘F.
d. Compressor air delivery pressure less than 5000 psig.
e. Compressor discharge capacity less than 20 cfm.

4.5.2.7 Precipitation.

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4. 5.2.7.1 Test procedure. The compressor unit shall be exposed to simulated rain by spraying fresh water at a minimum rate of 3 inches per hour directed simutaneously at all four sides and on top at an angle of approximately 45 degrees from the vertical. The apparatus for producing the spray shall emit water in the form of droplets rather than a fine mist. After 1/2 hour exposure to the spray, the compressor unit shall be started and shall deliver rated capacity for 1/2 hour, while the spray continues.

4.5.2.7.2 Failure criteria. Any one or more of the following shall constitute failure of this test

s. Inability of the compressor unit to start.
b. Faulty operation of any electrical switch or gage.
c. Leakage of water into internal component such as (but not limited to) the compressor unit, engine crankcase, and fuel tank.
d. Failure to deliver rated capacity.

4.5.2.8 Tilted position.

4.5 .2.8.1 Test procedure. Operate the compressor unit in four tilted positions, each of which sha ll be 15 degrees from the horizontal plane about the compressor transverse and longitudinal axes. The test shall be conducted operating the compressor unit for 1 hour at rated capacity in each of the four positions, in a sequence whereby the compressor unit is operated first in the position least likely to adversely affect the lubrication of fuel systems, and lastly in the position most likely to adversely affect these systems. The fuel tank shall be full at the start of each position run. Upon completing the last position run, stop the engine and allow the compressor unit to stand in this tilted position for 4 hours, then operate the compressor unit for one hour at rated capacity. Capacity measurement shall be made at least once during each of the five phases of the test.

4.5.2.8.2 Failure criteria. Inability of the compressor unit to produce rated capacity, evidence of compressor unit air or fluid leaks, or any malfunction shall constitute failure of this test.

4.5.2.9 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2. 1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc Increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.10 Air/gas purity test. The air purity and gas purity sampling test will be conducted in accordance with the procedures and requirements as stated in the publications and procedure documents at 3.24.

4.5.2.10.1 Failure criteria. Nonconformance to 3.10.3 or 3.23 or BB-A-1034, ASTM F 307, NAVMEDCOMINST 4260.3 and NAVSEA O994-LP-OO1-9O1O air testing procedures and results shall constitute failure of this test.

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4.5.2.11 Electromagnetic interference. The compressor unit shall be subjected to the test specified in MIL-STD-461 for class III C equipment. Nonconformance to MIL-STD-461 shall constitute failure of this test.

4.5.2.12 Slinging and tiedown provisions. The slinging and tiedown provisions shall be subjected to the loads and corresponding durations specified in MIL-STD-209 and the tiedown provisons shall be tested for the ultimate strength as specified in MIL-STD-209. Yield strength or ultimate strength not as specified in MIL-STD-209, shall constitute failure of this test.

4.5.2.13 Endurance phase. The compressor unit endurance test shall be conducted without shelter and at the climatic conditions existing at the place of test. The complete unit shall be operated (see 6.3.1) at rated capacity for 500 hours. Endurance test time will not be credited in less than 4-hour intervals of continuous operation, and a capacity computation shall be made from every fifth set of 4 hours data reading specified in 4.5.2.13.2. During the endurance phase the following shall be checked at least once.

a. Fuel tank capacity for conformance to 3.9.2.
b. Ease of starting the compressor unit with the filter system charged to a pressure not less than 3200 psig.

4. 5.2.13.1 Cycling phase. The cycling phase shall be for a total test time of 8 hours. The compressor shall be cycled to operate through a complete loading-unloading cycle every 5 minutes. Unloading shall be for not more than 1 minute duration. At the end of the cycling test phase, remove the compressor from cycling operation and operate the compressor at rated capacity for 1 hour. After 1-hour operation, make a capacity computation. During the cycling phase, observe the pressure control valve for operation in accordance with 3.11.

4.5.2.13.2 Test data. Air capacity shall be recorded as specified in 4.5.2.1 along with the constant factors and data used to compute the capacity figure. Air purity at the final filter outlet shall be recorded immediately before and after each filter change, and at least once in every 8 hours, plus or minus 0.5 hour of compressor operation. Air purity measurements shall be made by methods specified in BB-A-1034, ASTM F 307 and NAVCSEA 0994-LP-001-9020; by gas chromatography; or by any other means approved by the Government prior to test. Downtime data shall be recorded each time the compressor is stopped, including the stop-time and date, accumulated operating hours, start-time and date, duration of downtime in hours and minutes, the reason for each stop, and the amount of fuel and lubricants used. The following set of data shall be recorded every 4 hours, plus or minus 0.5 hour:

a. Air temperature, ambient.
b. Air temperature at compressing unit intake.
c. Air temperature at final filter inlet.
d. Air pressure at compressing unit intake.
e. Air pressure at final filter outlet.
f. Barometric pressure.
g. Engine rpm.

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4.5.2.13.3 Failure criteria. Any one or more of the following shall constitute failure of this test.

a. Inability of the compressor to produce rated capacity.
b. Inability of the compressor to operate for 500 hours without maintenance other than normal scheduled maintenance.
c. Fuel tank capacity not as specified in 3.9.2.
d. Pressure regulation not as specified in 3.11.
e. Air purity not as specified in 3.10.3 and 4.5.2.10.
f. Air temperature at final filter inlet not as specified in 3.10.2.

4.5.2.14 Reliability. Two compressor units shall be tested as follows:

4. 5.2.14.1 Test conditions. The operational and environmental test conditions shall be as follows:

a. Without shelter and at the climate conditions exiting at the place of test.
b. With the compressor units serviced in accordance with 4.5.2.1.
c. Shall operate at rated speed and capacity for not less than 1125 hours, with not less than 500 hours on a single unit.
d. Shall have capacity computations made on every 5th set of 4–hour data reading as specified in 4.5.2.1.
e. Test data shall be recorded as follows:

(1) Air capacity shall be calculated as specified in 4.5.2.1 along with constant factors and data used to compute the capacity figure.
(2) Air purity at the final filter in accordance with 4.5.2.10 recorded immediately before and after each filter change and at least once every 8 ±0.5 hours of compressor operation.
(3) Downtime data each time the compressor unit is stopped, to include: stop time and date, accumulated operating hours , restart time and date, duration of downtime in hours and minutes; reason, for each stop, and the amount of fuel and lubricants used.
(4) The following shall be recorded every 4 ±0.5 hours.

a. Air temperature, ambient.
b. Air temperature at compressor inlet.
c. Air temperature at final filter outlet.
d. Delivery pressure.
e. Air purity at compressor outlet.
f. Air purity at final filter outlet.

4.5.2.14.2 Accept/reject cirteria. The compressor units shall complete 1125 hours with one or less failures and not less than 500 hours on a single unit. A failure shall be described in 4.5.2.14.3 and all failure shall be determined by the Government. Figure 1 O-C curve applies.

4.5.2.14.3 Failure definition and scoring criteria. A failure for the purposes of this test is defined as any malfunction which the operator cannot

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remedy by adjustment, repair or replacement within 30 minutes (all waiting periods are excluded) utilizing the resources available to the parent unit when deployed and which causes any of the following:

a. Failure to commence operation, cessation of operation or degradation of performance below the designated levels.
b. Serious damage to the system.
c. Personnel safety hazards,
d. Air purity not as specified.

4.5.2.14.4 Test procedure. The units shall be operated continuously as described above except that preventative maintenance checks and services shall be made as prescribed by the equipment maintenance manuals. Only the maintenance established by the contractor in the maintenance schedule prior to test start is permitted during test. Stops for such maintenance shall be recorded as previously described.

4.5.2.15 Maintainability. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.6.2. Errors or inadequacies in the manual shall not be considered in assessing the maintenance ratio.

4.5.2.15.1 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.6.2 shall constitute failure of this test.

4.5.2.16 Operation.. Operate each compressor unit for not less than 10 hours utilizing all operator controls and determine capacity. Capacity may be measured by use of a calibrated meter or bottle pump method. Air purity shall be tested in accordance with 4.5.2.10 after initial 30 minutes, after 4 hours and prior to completion of the 10-hour operation test. Inability of each compressor unit to produce no less than 20 cfm at 5000 psi, while meeting air purity requirements or any malfunction, leakage or rough or irregular operation shall constitute failure of this test.

4.5.2.17 Pendulum-impact test. The compressor shall be subjected to the pendulum-impact test in accordance with test method No. 5012 of FED-STD-101, except the veritcal height of drop shall be 18 inches, and compressor will be unboxed and securely mounted to a platform which will extend beyond the frame or any member of the compressor on all four sides and designed to receive the full impact of the abutment.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined and tested in accordance with the quality assurance provisions of MIL-C-3600 and with the special requirement of section 5 herein.

5. PAGKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with MIL-C-3600 with the exception that interior surfaces of air receivers which will contain air used for breathing or respiratory

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purposes and located downstream from the air-oil separator shall not be preserved. The degree of preservation, packing, and marking shall be level A, level B, or commercial as specified (see 6.2).

6. NOTES

6.1 intended use. The compressor unit is intended to supply clean high pressure breathing air for charging SCUBA bottles, and air storage systems used for diving purposes.

6.2. Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number , and date of this specification.
b. Time frame required for submission of the first article units.
c. When the Government will conduct the first article model examination and tests.
d. Time interval required for submission of the electromagnetic interference report (see 3.14).
e. Color required when other than as specified (see 3.20).
f. When the welding process w1ll not be at the option of the contractor (see 3.21.3).
f. Degree of preservation, packing, and marking required (see 5.1).

6.3 Definitions.

6.3.1 Continually. operation in increments of not less than 3 hours except for interruptions required for normal servicing of the compressor unit. Downtime shall not be considered as Operation time.

6.3.2 Maintenance ratio (MR). The ratio of the number of maintenance man-hours required for both scheduled maintenance and unscheduled repairs (excluding operator daily checks and services), to the number of hours of operation.

6.4 First article. When a first article inspection is required, the items should be a production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.5 Approved for Navy use list. The contracting officer shall consult the Diving Equipment Authorized for Navy Use, 10 June 87 (NAVSEA Inst 10560.2) to Determine which vendors and equipment meet the safety standards for life support diving equipment as established by the Naval Sea Systems Command (NAVSEA), Alexandra, Virginia prior to open solicitation for this unit.

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6.6 Subject term (key word) listing.

20 CFM
5000 PSI
Compressor unit
Diesel engine driven
Diving Air
Life support

6.7 Notification of conflicts. The contracting officer will formally contact in writing the Belvoir, Research, Development, and Engineering Center, Fort Belvoir, VA., and request assistance to determine which document of procedure takes precedence (see 3.25).

6.8 Documentation. Documentation for cleaning procedures, and air purity sampling test procedures and results for the first article unit and each production unit shall be provided to the Government.

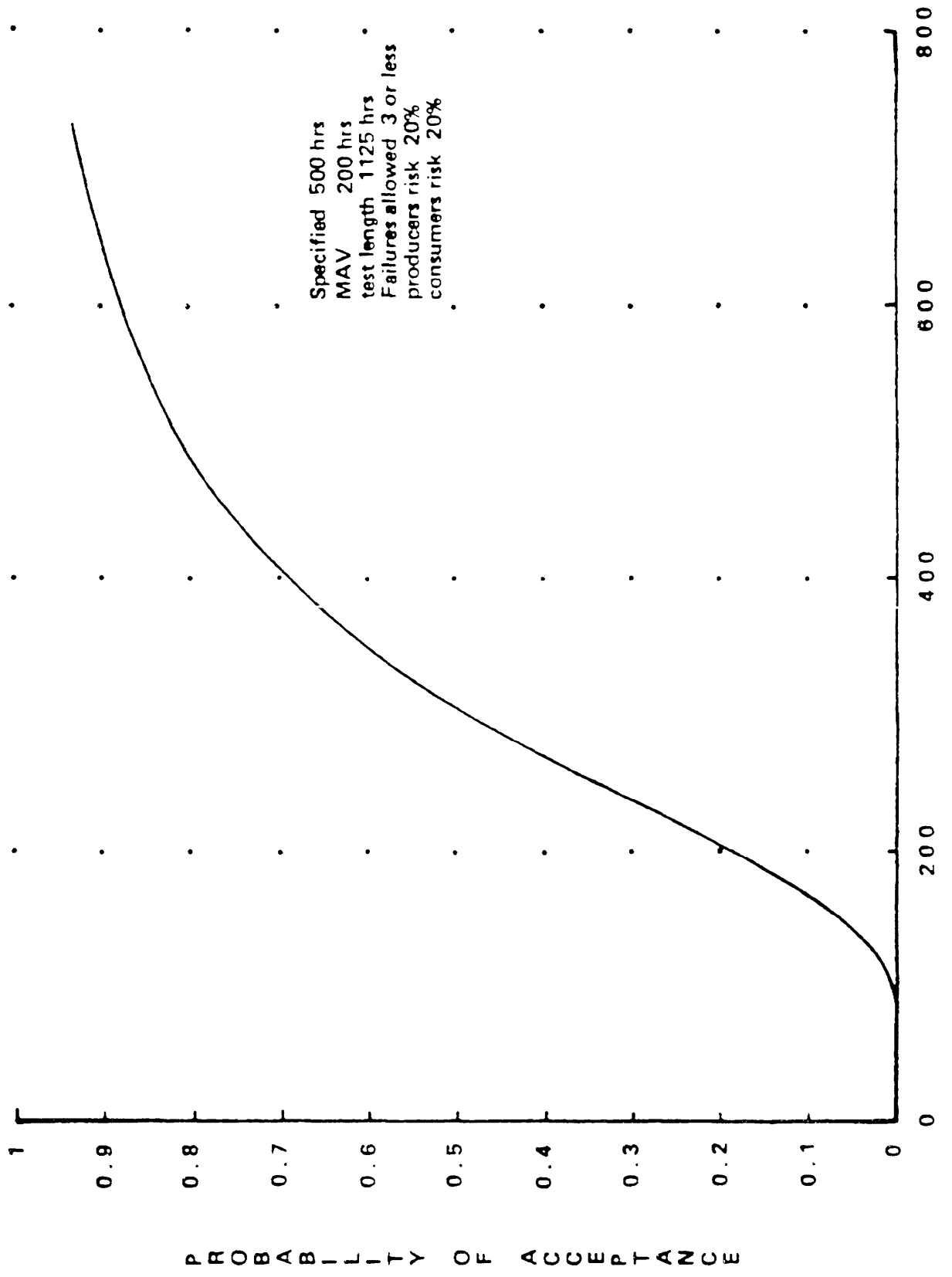
6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revison to identify changes with respect to the previous issue due to the extensiveness of the changes.

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X-4718

Figure 1. OPERATING CHARACTERISTIC (O-C) CURVE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52973A(ME)	Compressor Unit, 20 CFM, 5000

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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